Natural Origin III — The Enclosure Universe

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Abstract

This paper proposes an "enclosure universe" framework in which the only physical entity is the space-velocity field $\mathbf{g}(\mathbf{x}, t)$. The system introduces a single basic axiom—the composite light-speed axiom $|\mathbf{g}| = c$, i.e. the modulus of the composite motion of space is everywhere equal to the speed of light. As a result, all physical structures no longer rely on parallel assumptions of "particle–force–field", but are understood uniformly as stable solutions of one and the same space-velocity field at different scales and enclosure levels.

Under the strict condition $|\mathbf{g}| = c$, we derive the unique admissible helical decomposition. By introducing an allocation-angle function we ensure that any swirling structure can change only its direction but not the magnitude of the velocity, thereby eliminating the hidden superluminal inconsistency present in traditional helical models. The spatial structure is minimally and uniquely decomposed into three layers: a phase layer ϕ, a swirling layer $\Omega = |\nabla \times \mathbf{g}|$, and an enclosure layer $n = \frac{1}{c}\nabla \cdot \mathbf{g}$. This naturally defines three geometric emergent fields

$$\mathbf{E} = \nabla\phi, \quad \mathbf{B} = \nabla \times \mathbf{g}, \quad \mathbf{A}_g = c^2 \nabla n,$$

where $\mathbf{A}_g$ is, by definition, the acceleration field.

In intuitive directional terms, these three fields always satisfy a fixed orthogonal structure: the magnetic field $\mathbf{B}$ is chosen as the local axial direction (Z-axis), while the electric field $\mathbf{E}$ and the fall/gravitational field $\mathbf{A}_g$ lie in the plane perpendicular to it; they are mutually orthogonal and co-rotate in that plane as the phase (ϕ) evolves. This "vertical magnetic axis + in-plane orthogonal double field" geometry is preserved at microscopic, macroscopic, and extreme enclosure scales.

At the microscopic level, we construct enclosure models for the electron, proton and neutron. The electron is defined as the first stable enclosure state (CU-1); its minimal two-branch phase structure necessarily forms a closed right-handed double helix, from which we rigorously obtain eight topological event points as the minimal enclosure mother number. We then introduce the enclosure-consistency invariant $M\Omega = \mu_g c$, locking the mass scale and frequency scale into a single geometric constant system. The proton is shown to be the second stable enclosure level built upon the electron; its mass is uniquely determined by a second-order closure threshold and a second-order visibility threshold, and this naturally yields single-channel visibility and universality of charge. The neutron is modeled as a "phase sign-reversal compensation shell" that de-electrifies the enclosure; we give a topological proof of charge neutrality while preserving magnetism, and derive a calculable leading formula for the neutron–proton mass difference, consistent with MeV-scale observations.

At the field-structure level, the magnetic field is rigorously redefined as the enclosed flux tube of Ω. With the helical decomposition that enforces $|\mathbf{g}| = c$, we prove that magnetic flux tubes must

form, and that if they do not close they must appear in pairs as a jet port (N-pole) and an intake port (S-pole), thereby giving the geometric origin of the absence of magnetic monopoles and of the paired north/south poles. Stable flux tubes satisfy the dimensionless closure condition

$$\frac{\Omega}{c}\oint d\,s = 8\pi k$$

which generalizes the electron's 8-event mother number to a macroscopic 8k rhythm and clarifies the essential distinction between magnetic states (with ϕ locked) and radiative/light states (with ϕ released).

At extreme and macroscopic scales, we redefine black holes as the limiting state of two-dimensional spherical-shell enclosure, derive the 8π enclosure invariant and a no-singularity theorem, and interpret jets as boundary-port rupture of the enclosure. On the scale of the solar system, introducing a background rotation frequency $\Omega_{\mathrm{bg}}(r) = c/r$, we propose that planetary orbits are standing-wave nodes of 8k locked-phase enclosure, derive the shell rules $r \propto k^2$ and $T \propto k^3$, and perform an integer-shell audit using public planetary data, obtaining consistency at the $10^{-4} \sim 10^{-3}$ level.

In summary, this work completes a cross-scale enclosure-geometry closed loop from electrons, flux tubes, and black holes to the solar system. Within this framework the universe is neither a "particle universe" nor a "wave universe", but an enclosure universe: what we call particles, fields and interactions are all inevitable manifestations of the space-velocity field at different enclosure levels.

Chapter 1 Right-Handed Helical Flow: The Single Axiom and Three-Layer Structure of the Enclosure Universe

— Fixing the Definition Chain Before Discussing Electrons, Flux Tubes, Black Holes and Cosmic Scales

1.1 Starting point of this book: not particles and forces, but the motion of space itself

This book does not treat "particles, fields, interactions" as parallel fundamental objects. It recognizes only one physical entity:

The space-velocity field $g(\mathbf{x}, t)$: the state of motion of space itself at every point.

From this viewpoint, a "body" is not a third thing that occupies space, but a stable enclosure state of the space-velocity field at some scale.

Thus the first question this book addresses is not "how forces act", but the deeper question: what kinds of spatial motion structures can be locked, enclosed, and remain stable over long times?

To avoid any ambiguity later on, this chapter fixes once and for all the only definition chain allowed in the book. From here on, all subsequent chapters may only use the definitions established in this chapter; no new definitions will be introduced.

1.2 The single axiom: composite light-speed axiom $|\boldsymbol{g}| = \boldsymbol{c}$

Axiom 1 (Composite light-speed axiom)

$$|g(\mathbf{x}, t)| = c \tag{1.1}$$

This is not "some assumption about propagation speed", but a restriction on the spatial entity itself:

The modulus of the composite motion of space is equal to the speed of light at every point.

There is only one direct consequence, but it is crucial: any decomposition (radial / tangential / axial) must satisfy a Pythagorean constraint; otherwise the entire theory self-destructs immediately.

All later structures involving helices, swirling, flux tubes and black-hole shells are constrained by this axiom.

1.3 Correct helical decomposition: using an allocation angle to enforce $|\boldsymbol{g}| = \boldsymbol{c}$

In local cylindrical coordinates (r, θ, z), decompose the space-velocity field g into radial, tangential and axial components:

$$g = g_r\,\hat{\mathbf{r}} + g_\theta\,\hat{\theta} + g_z\,\hat{\mathbf{z}} \tag{1.3.1}$$

Because the only axiom of this book requires

$$|g| = c \tag{1.3.2}$$

any helical structure may change direction but must not change the velocity modulus.

To describe "swirl + advance" within a single expression, we must introduce an allocation-angle function $\chi(r, t)$ and write

$$g_\theta = c \sin\chi\,(r, t), \quad g_z = c \cos\chi\,(r, t), \quad g_r \to 0 \tag{1.3.3}$$

This immediately yields

$$g_\theta^2 + g_z^2 = c^2 \tag{1.3.4}$$

so, the composite light-speed axiom is satisfied exactly at all times and positions.

Important clarifications (to avoid misunderstanding):

- At certain enclosure boundaries (such as tangentially saturated shells or tube walls) one may choose g_z comparatively small;
- In more general structures (e.g. inside flux tubes or in the solar-system background flow), g_z is allowed to be an $O(c)$ component;
- In all cases, the magnitude of g_z is always constrained jointly by (1.3.3) and $|g| = c$.

1.4 Correct helical decomposition: use allocation angle to force $|g| = c$

In cylindrical coordinates (r, θ, z), write the space-velocity field as

$$g = g_r\,\hat{\mathbf{r}} + g_\theta\,\hat{\theta} + g_z\,\hat{\mathbf{z}} \tag{1.2}$$

To describe advance and swirl simultaneously without violating $|g| = c$, we again introduce the allocation angle $\chi(r,t)$ and set

$$g_\theta = c\sin\chi\,(r,t), \quad g_z = c\cos\chi\,(r,t), \quad g_r \ll c \tag{1.3}$$

Then

$$g_\theta^2 + g_z^2 = c^2 \tag{1.4}$$

holds automatically.

Forbidden form:
Setting $g_\theta = c$ and $g_z = c$ simultaneously would give $|g| = \sqrt{2}\,c$, directly violating the axiom.

The enclosure boundary condition shared by all stable enclosure states (electrons, flux-tube walls, black-hole shells) is

$$g_r \to 0, \quad g_\theta \to c \tag{1.5}$$

This is the tangential light-speed wrapping limit, which is:

- the geometric origin of single-channel visibility,
- the origin of the absence of magnetic monopoles, and
- the origin of the black-hole no-singularity result.

1.5 Three-layer structure $\phi - \Omega - n$: the minimal state space of space

This book does not introduce multiple "fields"; it recognizes only a three-layer decomposition of the same object g:

1.5.1 Phase / directional layer ϕ

$$\phi \in S^1 \tag{1.6}$$

ϕ is a dimensionless phase angle describing directional structure and topological winding. It is the geometric source of electric properties and information.

1.5.2 Swirl / structural layer Ω

$$\Omega \equiv |\nabla \times g| \tag{1.7}$$

Ω has dimension s^{-1}; it describes the strength of spatial swirling and is the source of magnetism and spin structure.

1.5.3 Enclosure / object layer n

The divergence $(\nabla \cdot g)$ has dimension s^{-1}.

We define the enclosure potential

$$n \equiv \frac{1}{c} \nabla \cdot g \tag{1.8}$$

n characterizes the scale of enclosure density and is the source of objecthood and the manifestation of mass.

1.6 Three geometric fields (with "dimension locking")

From the three-layer structure we naturally define three fields:

$$\mathbf{E}_g \equiv \nabla\phi \tag{1.9}$$

$$\mathbf{B}_g \equiv \nabla \times g \tag{1.10}$$

$$\mathbf{A}_g \equiv c^2 \nabla n \tag{1.11}$$

Note that (1.11) is the dimension-locking point:

- $n \sim \mathrm{s}^{-1} \Rightarrow \nabla n \sim \mathrm{m}^{-1}\mathrm{s}^{-1}$
- multiplying by c^2 yields an acceleration with units $\mathrm{m/s}^2$.

Thus the "gravitational / fall field" in this book is, by definition, an acceleration field, not an analogy.

1.7 Two immediate hard geometric theorems

Theorem 1 (No magnetic monopoles)

$$\nabla \cdot (\nabla \times g) = 0 \tag{1.12}$$

This is a vector identity, independent of dynamics. Any magnetic structure must either close on itself as a loop or appear with ports in pairs.

Theorem 2 (Ports must come in pairs)
If an enclosed flow tube does not close, two ports must appear:

- a jet port (N-pole): structure opens outward;
- an intake port (S-pole): structure rolls back inward.

Ports are not "sources", but topological defect points on the enclosure boundary. Magnetic poles, black-hole jets and flux-tube ends are isomorphic in this book.

1.8 Definition of enclosure states: what is a "body"?

Definition 1.1 (Enclosure state)

If the space-velocity field forms a stable boundary in some region such that

- the radial component is suppressed: $g_r \to 0$;
- the tangential component saturates: $g_\theta \to c$;
- the three layers (ϕ, Ω, n) can be locked simultaneously;

then that region corresponds to a stable enclosure state (a body).

This definition applies uniformly to

- electrons (minimal enclosure),
- protons / neutrons (higher-order enclosures),
- flux tubes (Ω-enclosed flow tubes),
- black holes (spherical-shell extreme enclosures),
- the solar system (macroscopic standing-wave enclosure).

1.9 Chapter summary and roadmap of the book

This chapter has fixed the only structures allowed throughout the book:

- single axiom: $|g| = c$;
- directional convention: Z-axis sets the swirl axis, the X–Y plane sets phase and enclosure;
- correct helical decomposition: allocation angle χ;
- three-layer structure: ϕ, Ω, n;
- three field definitions: $\nabla\phi, \nabla \times g, c^2\nabla n$;
- hard geometric theorems: no magnetic monopoles, ports must come in pairs;
- definition of a body: stable enclosure state.

Subsequent chapters are arranged as follows:

- Chapter 2: the electron and the 8-event topological mother number;
- Chapter 3: the proton and the geometric derivation of mass;
- Chapter 4: the neutron and ϕ-compensation enclosure;
- Chapter 5: flux tubes, jet ports (N-poles) / intake ports (S-poles) and the 8k rhythm;
- Chapters 6–9: the frequency tree, black holes and the solar system;
- Chapter 10: the interface to atoms and elements.

Chapter 2 Electron: The First Stable State (CU-1) in an Enclosure Universe

— Double-Helix Closure, the 8-Event Mother Number, and the Enclosure Consistency Invariant

2.1 Ontological statement: the electron is not a point particle but a stable enclosure state

In Chapter 1 we gave the criterion for an enclosure state:
when the space-velocity field g forms a tangential light-speed wrapping in some region and maintains $|g| = c$, that region constitutes a stable enclosure state.

Definition 2.1 (Electron)
The electron is the first stable enclosure solution of the space-velocity field g, denoted CU-1.

This definition implies:

- The electron is not a point;
- The electron is not a probability cloud;
- The electron is not "a moving thing".

The electron is a confined, locked geometric state.

2.2 Boundary condition of electron enclosure: tangential light-speed wrapping

Using the correct helical decomposition from Chapter 1, on the electron's enclosure boundary we have

$$g_r \to 0, \quad g_\theta = c \sin \chi\,(s), \;\; g_z = c \cos \chi\,(s), \tag{2.1}$$

where s is the arc-length parameter along the enclosure path.

Thus we automatically obtain

$$g_\theta^2 + g_z^2 = c^2 \tag{2.2}$$

Key clarification:

The "helix" of the electron is not an oscillation in speed magnitude, but a continuous rotation of direction under the constraint of a fixed speed modulus.
The helix is a directional structure, not an amplitude structure.

2.3 Two-branch phase structure: the origin of the double helix

In this framework, the electron's electric property arises from the topological winding of the phase layer ϕ. The minimal nonzero winding magnitude is $|\Delta\phi| = 1$.

To form a stable yet not over-constrained structure on the enclosure boundary, the electron adopts a minimal two-branch phase structure:

$$\phi_2 = \phi_1 + \pi \tag{2.3}$$

Definition 2.2 (Electron double helix)
The electron enclosure state is formed by two phase branches $\phi = +\pi$ and $\phi = -\pi$ locked on the same enclosure boundary. These two branches swirl right-handedly around the Z-axis in space and jointly form a closed double-helix enclosure structure.

2.4 The 8-event topological mother number: the inevitable outcome of minimal closure

This is one of the most central geometric facts of the entire work.

2.4.1 Two conditions required for closure

For the electron enclosure to be stable, it must simultaneously satisfy:

- Condition C1 (phase closure): after one circuit along the enclosure path, the phase relations must return to their initial state;
- Condition C2 (directional closure): the helical directional structure determined by Ω must close self-consistently.

Because the two branches differ by π, a single 2π circuit cannot satisfy both conditions at once.

2.4.2 Two turns and the minimal event count

To bring the "double branch + directional structure" back to the initial state simultaneously, the enclosure path must run twice around (total angle 4π).

In each circuit, the helical direction rotates continuously and passes through four unavoidable directional-quadrant event points.

Thus the minimal number of closure events is

$$N_{event} = 2 \times 4 = 8 \tag{2.4}$$

Theorem 2.1 (8-event topological mother number)
As the minimal two-branch enclosure state, a stable closure of the electron necessarily contains eight topological event points.
"8" is the minimal closure rhythm of the enclosure universe. This is not a numerical assumption but a geometric necessity.

From a more formal perspective, the two-branch phase structure requires the directional field on the minimal closed loop to realize a nontrivial double-valued mapping; its minimal closure multiplicity is 2. Combined with the two branches and two turns of the path, this locks in a minimal topological event count of 8.

2.5 Enclosure consistency invariant: locking mass and rotation frequency

In this framework, the mass of an enclosure state is given by its enclosure "rotation length":

$$M = \mu_g \ell \tag{2.5}$$

The frequency chain gives the rotation-frequency scale:

$$\Omega \sim \frac{c}{\ell} \tag{2.6}$$

Combining the two yields

$$M\Omega \sim \mu_g c$$

In this book, this relation is elevated to a hard constraint for enclosure states:

$$\boxed{M\Omega = \mu_g c} \tag{2.7}$$

Theorem 2.2 (Enclosure consistency invariant)
For any stable enclosure state, the product of its mass scale and rotation-frequency scale is uniquely fixed by $\mu_g c$.

This invariant will be used in later chapters to

- derive charge universality (Chapter 3), and
- derive the black-hole 8π invariant (Chapter 7).

2.6 Decoupling and recoupling of the double helix: a crucial distinction of levels

This is key to avoiding misunderstandings.

2.6.1 Strict separation of core and shell

Definition 2.3 (Core–shell separation)

- Electron core: the stable enclosure state (CU-1), which does not split and does not disintegrate;
- Electron shell: the $\phi - \Omega$ directional structure outside, which can unfold, interfere and propagate.

2.6.2 Where does "decoupling" occur?

The "decoupling" of the double helix occurs only in the shell layer, as magnetic flux tubes, and never in the enclosure core.

- The shell phase can be perturbed by boundary conditions;
- It can form interference patterns, superpositions and propagating states;
- The core remains a closed enclosure at all times.

2.6.3 What does recoupling mean?

Recoupling does not mean "recreating the electron"; it means that the shell phase is re-locked to the enclosure boundary.

This distinction will be used directly for:

- the correct explanation of double-slit interference;
- magnetic–optical conversion (Chapter 5); and
- the geometric interpretation of measurement and "collapse".

2.7 Chapter summary

This chapter has established four unavoidable hard conclusions about the electron model:

1. The electron is the first stable enclosure state (CU-1), satisfying tangential light-speed wrapping and $|g| = c$;
2. The electron has a two-branch phase structure, forming a right-handed closed double helix;
3. Minimal stable closure necessarily produces the 8-event topological mother number;
4. The electron obeys the enclosure consistency invariant $M\Omega = \mu_g c$.

Transition to the next chapter

The next chapter will show that the proton is not another fundamental particle, but the second stable enclosure level built on top of the electron's double helix.

Its mass will no longer be "input", but will be strictly derived from a second-order closure threshold π^2 and a second-order visibility threshold α^2, and this will naturally yield single-channel visibility and charge universality.

Chapter 3 Proton: The Second Stable Enclosure State above the Electron Double Helix

— Geometric Derivation of Mass, Double Thresholds, and Single-Channel Visibility

3.1 Existence before properties: why the proton "must" exist

In the previous chapter we showed that:

- The electron (CU-1) is the minimal stable enclosure state;
- Its stability arises from double-helix closure and the 8-event topological mother number;
- The electron satisfies the enclosure consistency invariant $M\Omega = \mu_g c$.

But electrons alone are not enough to build the material world. The reason is not "insufficient

interaction strength", but insufficient enclosure hierarchy:

- The electron primarily provides an external manifestation of the ϕ layer (direction / information);
- Without a stronger n −layer core, shell structures cannot form stable residence at larger scales;
- Atomic nuclei, elements and macroscopic structures cannot emerge.

Conclusion:
If an enclosure universe is to move from a "single stable state" to "extendable material structures", a second stable enclosure level must appear.

This second level is not "one more particle", but a jump in enclosure depth.

This chapter will show that the proton is the unique minimal realization of that jump.

3.2 Two unavoidable existence thresholds: π^2 and α^2

The second stable enclosure state is "heavier" not because of stacking, but because it crosses thresholds.

3.2.1 Second closure threshold: π^2 (closure of a closure)

The electron is the shortest first-order closure: completing one minimal geometric loop is sufficient for stability.

If the proton were merely a "larger electron", it would not remain stable over long times.

The second stable state must satisfy a stronger condition: the closure itself must be closed again.

The smallest purely geometric cost, expressed as a dimensionless factor, can only be π^2. This is not a fit, but the minimal cost of "second complete coherence over the full circle".

3.2.2 Second-order visibility threshold: α^2 (visibility of visibility)

As the minimal enclosure state, the electron does not itself require a macroscopic visibility threshold.

But for a second stable basic object to be "seen" at macroscopic electromagnetic scales, its enclosure residual must cross a second-order visibility threshold:

- α: the minimal visible fraction of a single enclosure outlet in the macroscopic layer;
- α^2: the squared visibility threshold that determines whether a new stable level can appear.

Physical meaning:
α^2 is not "injecting an electromagnetic constant"; it is the existence condition for the second stable object.

3.3 Leading formula for the proton mass: direct derivation from existence conditions

Relative to the first stable state, the second stable enclosure must pay both costs:

- The second closure threshold π^2;
- The second-order visibility threshold α^2.

Thus, the existence condition can be written as

$$\frac{m_p}{m_e} \approx \frac{1}{\pi^2 \alpha^2} \tag{3.1}$$

This is not fitted from experiment; rather:

If this relation is not satisfied, the second stable level either does not exist or is unstable.

3.3.1 Numerical audit (discriminative, not a fit)

Taking $\alpha^{-1} \approx 137$, we find

$$\frac{1}{\pi^2 \alpha^2} \approx 1.90 \times 10^3.$$

The experimental ratio is $m_p/m_e \approx 1836$, differing by about 3–4%.

We do not treat this difference as an "error" to be hidden; instead we identify it as

$$\frac{m_p}{m_e} = \frac{\eta_\Omega}{\pi^2 \alpha^2}, \quad \eta_\Omega \simeq 0.96 \tag{3.2}$$

where η_Ω is a rotational-consistency correction in the second enclosure boundary layer, arising from stability rather than free fitting.

3.3.2 Ontological origin of the rotational-consistency correction: co-rotation mismatch between proton and background

In §3.3.1, the existence-threshold theorem gave the leading expression for the proton mass:

$$\frac{m_p}{m_e} \approx \frac{1}{\alpha^2 \pi^2} \quad (3.1)$$

The measured mass ratio differs from this by about 3–4%. This difference is not treated as numerical error or fitting residual; instead it is strictly interpreted as a rotational-consistency correction in the second enclosure boundary layer.

We introduce a dimensionless correction factor η_Ω and write the proton mass as

$$\frac{m_p}{m_e} = \frac{\eta_\Omega}{\alpha^2 \pi^2}, \quad 0 < \eta_\Omega < 1 \quad (3.2)$$

(1) Back-calculation and reproducible value of η_Ω

From (3.2) we can back-calculate

$$\eta_\Omega = \frac{(m_p/m_e)exp}{1/(\alpha^2\pi^2)} = (m_p/m_e)exp\,\alpha^2\pi^2 \tag{3.3}$$

With $\alpha^{-1} \approx 137$, one finds $\alpha^2\pi^2 \approx 5.26 \times 10^{-4}$.
Using $(m_p/m_e)exp \approx 1836.15$, we obtain

$$\eta\Omega \approx 0.9655, \quad 1 - \eta_\Omega \approx 3.45\% \tag{3.4}$$

This matches the repeatedly mentioned "about 3–4%" difference in the main text, and the back-calculation introduces no new free parameters.

(2) Physical meaning: co-rotation mismatch factor

In the SUF framework, η_Ω is no longer interpreted as a statistical or coherence factor, but is explicitly identified as a co-rotation mismatch factor.

Let

- Ω_p: the effective enclosure rotation frequency of the proton's second-level enclosure boundary in its stable state;
- $\Omega_{\mathrm{b}g}$: the local reference rotation frequency provided by the background right-hand helical flow at the scale of this boundary layer.

In the ideal limit, if the proton's enclosure boundary layer were exactly co-rotating with the background flow, we would have:

$$\Omega_p = \Omega_{\mathrm{b}g}, \quad \eta_\Omega = 1.$$

In reality, as a second stable enclosure state, the proton's boundary layer inevitably exhibits a small mismatch in rotation:

$$\Omega_p \neq \Omega_{\mathrm{b}g}$$

This mismatch compresses the ideal mass ratio predicted by "second closure + second-order visibility threshold", and naturally introduces the correction factor. We therefore define

$$\eta_\Omega \equiv \frac{\Omega_{\mathrm{b}g}}{\Omega_p} \tag{3.5}$$

as the minimal definition of the rotational-consistency correction.

Thus the 3.45% mass difference is, at the ontological level, equivalent to

$$\frac{\Omega_p - \Omega_{\mathrm{b}g}}{\Omega_p} = 1 - \eta_\Omega \approx 3.45\% \tag{3.6}$$

(3) Why this correction appears for the proton but not for the electron

For the electron as the first stable enclosure state, its minimal closure is rigidly locked by the "8-event mother number". At this level there is no independent second-enclosure boundary layer, so there is no resolvable co-rotation degree of freedom between the electron's enclosure rotation and the background swirling reference.

The proton is different. As a second stable enclosure level, its external stability is determined by a

second enclosure boundary layer that must satisfy both the enclosure conditions and stable matching with the background swirling reference at that scale. A rotational-consistency correction is therefore unavoidable, entering the mass formula via $\eta_\Omega < 1$, and this is the physical origin of the 3–4% systematic difference observed in experiment.

3.4 Single-channel visibility: a consequence, not a premise

We now introduce the proton's external electric property.

3.4.1 Why only one outlet can be visible

As a second stable enclosure state, the proton's internal $\Omega - \phi$ texture has a complexity of the same order as the mass ratio ($\sim 10^3$).

However, the enclosure boundary tends toward tangential light-speed wrapping; the consequence of energy minimization is:

- Multiple outlets ⇒ over-constrained phase conditions ⇒instability;
- Stable state ⇒ external degrees of freedom must be compressed.

Single-channel visibility theorem (proton)
At the macroscopic level, the second stable enclosure state can retain at most one topologically protected outlet.

Hence,

$$|\Delta\phi| = 1 \quad \Rightarrow \quad q_p = +e$$

3.4.2 Charge universality as a corollary

From the enclosure consistency invariant (Chapter 2),

$$M\Omega = \mu_g c$$

we define the unit outlet charge

$$q_0 \equiv k_{mq}\, \mu_g c$$

Charge readout is then

$$q = \Delta\phi\, q_0 \tag{3.3}$$

Both electron and proton satisfy $|\Delta\phi| = 1$, so

$$|q_e| = |q_p| = q_0$$

Charge universality is therefore not a hypothesis, but the shared outcome of the "minimal nonzero topological outlet".

3.5 Internal 8k modes and the "visible area" interface

Although only a single outlet appears externally, the proton internally retains higher-order enclosure rhythms with mode number

$$k \sim \frac{m_p}{m_e} \sim 10^3$$

These 8k-level modes are angularly averaged, leaving an effective coupling only in the outlet residual region.

The equivalent interface between the macroscopic world and the nucleus can be written as

$$A_{vis} = A_\Omega \, k_{shape} \, k_{coh}, \qquad A_\Omega = \pi \ell_\Omega^2 \tag{3.4}$$

This interface will be used later for:

- Chapter 5: the nuclear origin of flux tubes;
- The projection of nuclear outlets in atomic and hyperfine structure.

3.6 Chapter summary

This chapter has established the proton model at the ontological level. The conclusions must appear in the following order:

1. The proton is the second stable enclosure level, not another fundamental particle;
2. Its mass is uniquely derived from existence thresholds:

$$\frac{m_p}{m_e} = \frac{\eta_\Omega}{\pi^2 \alpha^2}$$

3. Single-channel visibility is the external manifestation of the second enclosure state;
4. Charge universality arises from the common readout of the minimal topological outlet;
5. Internal 8k modes couple to the macroscopic world through the “visible area” interface.

Transition to the next chapter

The next chapter will introduce the neutron. The neutron is not “a proton without charge”, but a de-electrified enclosure state with a ϕ-compensation shell. We will provide:

- A topological proof of charge neutrality via cancellation of winding numbers;
- A geometric explanation of why the neutron still has a magnetic moment;
- A calculable leading formula (MeV scale) for the neutron–proton mass difference.

Chapter 4 Neutron: A De-Electrified Enclosure with a ϕ-Compensation Shell

— Structural Definition, Charge Neutrality, Magnetic Moment Retention, and Geometric Calculation of the Mass Difference

4.1 What is a neutron: proton core + ϕ-compensation shell

In Chapter 3, the proton was established as the second stable enclosure state, with two key external conclusions:

1. Single-channel visibility: only one topological outlet appears externally;
2. Charge readout rule: $q = \Delta\phi\, q_0$.

On this basis, we now give the ontological definition of the neutron.

Definition 4.1 (Neutron)
A neutron is a de-electrified stable enclosure, whose structure consists of two parts:

(i) A proton enclosure core (retaining strong n-enclosure and Ω-texture, tending to $\Delta\phi = +1$);
(ii) A ϕ-compensation shell (a thin shell of opposite phase, carrying $\Delta\phi = -1$).

This compensation shell is not an "extra particle" but a phase reconfiguration of the enclosure boundary. It acts only in the outlet neighborhood and does not destroy the stability of the core enclosure.

4.2 Hard proof of charge neutrality: cancellation of topological winding numbers

We reuse the charge readout formula from Chapter 3:

$$q = \Delta\phi\, q_0 \tag{4.1}$$

The topological winding number of the proton core is

$$\Delta\phi_{core} = +1 \tag{4.2}$$

and that of the compensation shell is

$$\Delta\phi_{shell} = -1 \tag{4.3}$$

Hence the total winding of the neutron is

$$\Delta\phi_n = \Delta\phi_{core} + \Delta\phi_{shell} = 0 \tag{4.4}$$

Substituting into (4.1) gives

$$q_n = 0 \tag{4.5}$$

Conclusion:
A neutron is not "without electric structure"; its electric structure is fully compensated and closed topologically.

4.3 Why the neutron still has a magnetic moment: Ω does not vanish with ϕ-compensation

In the three-layer structure of this book, the three physical manifestations are strictly separated:

- Electric properties $\Leftrightarrow \nabla\phi$;

- Magnetic properties $\Leftrightarrow \Omega = |\nabla \times g|$;
- Objecthood / mass $\Leftrightarrow n$.

Compensating ϕ only affects the electric readout; it does not require Ω to vanish. Therefore,

$$q_n = 0 / \Rightarrow \Omega_n = 0 \tag{4.6}$$

More concretely:

- The Ω-texture of the proton core remains;
- The ϕ-compensation shell primarily rewrites the outlet phase;
- The core swirling structure is not erased.

Conclusion:
Electric neutrality and magnetism are not intrinsically opposed. A neutron can be electrically neutral yet retain a nonzero magnetic moment.

This will serve as an important interface in Chapter 5 for the link
"nuclear Ω-texture → macroscopic flux tubes."

4.4 Geometric origin of the neutron–proton mass difference: enclosure reconstruction energy of the compensation shell

We now reach the core question of this chapter:
Where does the mass difference come from, and how do we calculate it?

4.4.1 Mass and enclosure rotation length

In this framework, the geometric relations between mass and enclosure rotation length are

$$M = \mu_g \, \ell, \quad E = \mu_g c^2 \, \ell \tag{4.7}$$

Compared to the proton, the neutron has one additional ϕ-compensation shell, corresponding to an extra enclosure rotation length $\Delta\ell$:

$$m_n = m_p + \Delta m, \quad \Delta m = \mu_g \, \Delta\ell \tag{4.8}$$

The problem is thus reduced to: how to estimate $\Delta\ell$.

4.4.2 Minimal cost of the compensation shell: counted via the "outlet residual"

The compensation shell acts only in the outlet region rather than throughout the entire proton enclosure. Its energy cost must therefore obey two requirements:

1. Proportionality: it depends only on the outlet residual;
2. Geometric character: it is set by the enclosure tilt angle and the symmetry of the layers.

In Chapter 3, α was identified as the minimal scale of macroscopic visibility. The minimal relative cost of the compensation shell should thus be proportional to α.

In addition:

- The minimal enclosure tilt angle of the electron's double helix is $\theta = 30°$;
- The effective projection factor is $\sin^2\theta$;
- In the three-layer orthogonal structure ($\phi \perp \Omega \perp n$), only 3/4 of the components participate effectively in reconstruction.

We therefore obtain the minimal relative increment

$$\frac{\Delta m}{m_p} = \frac{3}{4}\,\alpha\,\sin^2\theta \tag{4.9}$$

4.4.3 Leading formula and numerical substitution (discriminative)

Take

$$\alpha \simeq \frac{1}{137}, \quad \theta = 30° \Rightarrow \sin^2\theta = \frac{1}{4}$$

Then

$$\frac{\Delta m}{m_p} = \frac{3}{16}\,\alpha \approx 0.00137 \tag{4.10}$$

The corresponding energy difference is

$$\Delta E = (m_n - m_p)c^2 \approx 0.00137 \times 938\,MeV \approx 1.28\,MeV \tag{4.11}$$

This agrees with the observed value $1.293\,MeV$ at the 1% level, and crucially, the neutron mass was not used as an input.

Conclusion:
The neutron–proton mass difference is the enclosure reconstruction energy introduced by the compensation shell in the outlet residual region.

4.5 Two direct corollaries

Corollary 1: Neutrons are stable in nuclei but unstable when free

Since Δm arises from boundary-layer reconstruction energy, this reconstruction can be stabilized inside the nucleus (where enclosure boundary conditions are constrained), whereas in free space the compensation shell decouples more easily, making the neutron unstable.

Corollary 2: Minimal structural channel of β decay

When the compensation shell can no longer remain closed, its minimal release path is the decoupling and reconstruction of the ϕ-compensation shell, not the introduction of a new fundamental interaction.

In this framework, the weak interaction is re-located as a geometric channel of enclosure breakdown.

4.6 Chapter summary

This chapter has established four unavoidable hard conclusions for the neutron model:

1. The neutron is a de-electrified enclosure: proton core + ϕ-compensation shell;
2. Electric neutrality arises from cancellation of topological winding numbers: $\Delta\phi_n = 0$;
3. Magnetism comes from preserved Ω: neutrality does not mean lack of structure;
4. The leading mass-difference formula $m_n = m_p\left(1 + \frac{3}{4}\alpha \sin^2\theta\right), \quad \theta = 30°$, gives a MeV-scale, discriminative prediction.

Transition to the next chapter

The next chapter turns to the field-structure layer: the magnetic field is not "lines of force", but an enclosure flow tube of Ω. We will systematically answer:

- Why flux tubes must form;
- Why jets (N poles) and intakes (S poles) necessarily appear in pairs;
- How the electron's 8-event mother number lifts to a macroscopic 8k rhythm.

Chapter 5 Magnetic Field and Flux Tubes: Geometry and Dynamics of Ω-Enclosure Flow Tubes

— Jets (N Poles), Intakes (S Poles), the 8k Rhythm, and Double-Helix Decoupling–Recoupling

5.1 Main conclusion first: the magnetic field is not "lines of force" but an enclosure state

Within this framework, the ontological definition of the magnetic field can be stated in one sentence: The magnetic field is the enclosure state of the swirling layer Ω of the space-velocity field g in free space.

Mathematically this is equivalent to

$$\mathbf{B}_g \equiv \nabla \times g, \quad \Omega \equiv |\nabla \times g| \tag{5.1}$$

Therefore:

- The magnetic field is not a set of drawn lines;
- Magnetic flux is not an abstract flux;
- A magnetic structure is a bundle of genuinely existing Ω-enclosure flow tubes—flux tubes.

5.2 Direction structure first: division of roles between the Z-axis and the X–Y plane

Continuing the directional convention from Chapter 1:

- The Z-axis is the main rotation-frequency axis of Ω (flux-tube axis);
- The X–Y plane is where the phase ϕ and enclosure gradient n live.

Thus:

- Flux tubes extend along the Z-axis;
- The outward jet (N pole) and inward intake (S pole) appear in the X–Y plane;
- Any "direction of a magnetic field line" is essentially the spatial main axis of $(\nabla \times g)_z$.

5.3 Minimal structural hypothesis for flux tubes (required for enclosure)

Consider a flux tube along the Z-axis in cylindrical coordinates (r, θ, z). The minimal—and only allowed—assumptions are:

1. Axial symmetry: $\partial_\theta(\cdot) = 0$;
2. Steady state: $\partial_t(\cdot) \approx 0$;
3. No radial leakage in the core: $g_r = 0$.

Thus

$$g = g_\theta(r)\,\hat{\theta} + g_z(r)\,\hat{z} \tag{5.2}$$

5.4 Why flux tubes must form: existence proof

Flux tubes are not empirical constructs; they are self-organized enclosure solutions of Ω under the constraint $|g| = c$.

To prove their existence, consider the minimal energy functional on the tube cross-section:

$$\mathcal{F}[\chi] \int r\ dr[|\nabla \times g|^2 + a(\chi')^2 + u|\nabla \times g|^4] \tag{5.2.1}$$

where $\chi(r)$ is the helical allocation angle, and $a, u > 0$ are positive weights.

Varying χ $(\delta\mathcal{F}/\delta\chi = 0)$ gives an equation of the form

$$\chi'' + \frac{1}{r}\chi' - \frac{\sin\chi\cos\chi}{r^2} + \cdots = 0 \tag{5.2.2}$$

The structural meaning:

- The first two terms enforce cross-sectional smoothness;
- The third term prevents χ from collapsing to 0 or diverging to $\pi/2$;

- Stable stationary points necessarily correspond to finite-radius tube solutions.

About the coefficients.
The coefficients a, u in (5.2.1) only ensure the functional is positive and tube solutions exist. All subsequent results—8k closure, enclosure invariants, frequency scaling—do not depend on their numerical values. They are therefore not new free parameters in any observable prediction.

To describe "swirl + forward motion" without violating $|g| = c$, we must write

$$g_\theta(r) = c \sin\chi(r), \quad g_z(r) = c \cos\chi(r), \quad |g| = c \tag{5.3}$$

Key reminder:
Any ad hoc form like $g_\theta \sim cf(r)$, $g_z \sim ch(r)$ that does not introduce $\chi(r)$ will inevitably produce $|g| > c$ somewhere and self-destruct the theory.

5.5 Where do "field lines" come from? Explicit curl calculation

Under axial symmetry with $g_r = 0$, the curl components are

$$(\nabla \times g)z = \frac{1}{r}\frac{d}{dr}(rg\theta), \quad (\nabla \times g)_\theta = -\frac{dg_z}{dr} \tag{5.4}$$

Substituting (5.3), we obtain

$$(\mathbf{B}_g)z = c\left(\frac{\sin\chi}{r} + \chi'(r)\cos\chi\right) \tag{5.5}$$

$$(\mathbf{B}_g)\theta = c\,\chi'(r)\sin\chi \tag{5.6}$$

Conclusion (ontology of magnetic field lines):
A "magnetic field line" is not literally a line, but the direction of the main axis of $(\nabla \times g)_z$.

5.6 Closure condition for flux tubes: dimensionless phase accumulation and the 8k rhythm

As Ω-enclosure flow tubes, flux-tube stability must be characterized by a dimensionless closure phase, not by the magnitude of a line integral. This is dictated jointly by dimensional consistency and enclosure geometry.

Since Ω has units of s^{-1}, while phase must be dimensionless, we define the enclosure phase on the tube cross-section as

$$\Delta\Phi_{tube} \equiv \oint \frac{\Omega}{c}\, ds \tag{5.6.1}$$

where ds is the arclength element along a closed loop.

Explanation: Ω/c has dimension m^{-1} and is precisely the "phase density along the path". Using $\oint \Omega\, ds$ directly would give incorrect dimensions and is strictly forbidden here.

Stable enclosure demands that this phase be an integer multiple of the minimal topological closure rhythm, hence

$$\oint \frac{\Omega}{c}\, ds = 8\pi k, \qquad k \in \mathbb{Z}^{+} \tag{5.6.2}$$

Here:

- The factor 8 comes from the electron's minimal topological mother number (Chapter 2);
- k labels multi-layer or multi-mode enclosure levels.

This condition explains why 8, 16, 32, … rhythms are ubiquitous in macroscopic magnetic structures, without any new free parameters.

5.7 Jets (N poles) and intakes (S poles): not empirical, but topologically necessary

This is the central point of the chapter.

5.7.1 Geometric origin of "no magnetic monopoles"

$$\nabla \cdot (\nabla \times g) = 0 \tag{5.7}$$

is a vector identity, independent of dynamics.

Thus:

- Ω-flow tube must either close on itself; or
- Terminate in pairs of endpoints.

5.7.2 Geometric definitions of jets (N) and intakes (S)

- Jet (N pole): local break of the enclosure boundary where the Ω-structure flips outward;
- Intake (S pole): where the Ω-structure rolls back in from outside and re-encloses.

Endpoints are not "sources" or "sinks"; they are topological defects of the enclosure flow tube. This gives a unified description of bar-magnet poles, flux-tube ends, and black-hole jets.

Boundary between magnetic and radiative states: closure of ϕ

A flux tube corresponds to a stable enclosure state of Ω when the phase layer ϕ is locked. Its key feature is

$$\nabla\phi \cdot (\nabla \times g) = 0 \tag{5.8}$$

i.e., the phase layer ϕ is locked in a closed configuration and does not participate in propagation.

When external conditions prevent ϕ from remaining closed and introduce time dependence, the enclosure condition breaks down and the system undergoes a state transition:

$$\nabla\phi \cdot (\nabla \times g) \neq 0 \;\Rightarrow\; \text{propagating state} \tag{5.9}$$

Thus:

- Magnetic state: ϕ locked + Ω enclosed;
- Photon state: ϕ released + Ω propagating.

This criterion is consistent with later discussions of stationary versus propagating states, and does not rely on extra empirical formulas.

5.8 The 8k rhythm: from the electron's 8 mother events to flux-tube "quantization"

Stable closure of a flux tube must be characterized by dimensionless phase accumulation. Define

$$\Delta\Phi_{tube} \equiv \oint \frac{\Omega}{c}\, ds \tag{5.10}$$

Stable enclosure requires

$$\Delta\Phi_{tube} = 8\pi k, \qquad k \in \mathbb{Z}^{+} \tag{5.11}$$

Thus:

- Segmentation of ends into 8, 16, 32, … sectors;
- Cross-sectional patterns of flux tubes;
- Preferred orders of multipole structures

all originate from the electron's 8-event mother number.

5.9 Structure of jets (N) and intakes (S) in static flux tubes, double-helix decoupling, and the origin of dynamic magnetic wave packets

This section gives a clear statement of the internal structure and endpoints of static flux tubes, clarifying the physical meaning of jets (N poles) and intakes (S poles) and how the double-helix structure decouples and re-encloses at the tube ends. It also emphasizes: only when the flux tube as a whole loses enclosure and disintegrates into propagating packets do electromagnetic waves (dynamic magnetism) arise.

5.9.1 Double-helix enclosure structure of a static flux tube

In a steady magnetostatic state, the magnetic field does not consist of independently propagating lines; it exists in the form of flux tubes. A flux tube is an enclosure of a swirling structure of space, with a definite double-helix structure inside:

- The double helix is not two independent channels, but two phase branches of the same swirling channel;
- Under steady conditions, enclosure constraints compress them into a single channel with a fixed relative relation;
- The double-helix core exists and remains continuous throughout the static magnetic state; it never disintegrates.

Thus static magnetism is not a “stationary wave”, but a stable enclosure object with internal swirl and phase structure.

5.9.2 Jets (N) and intakes (S): endpoints of static flux tubes

Geometrically, a static flux tube must have two endpoints, called the jet (N pole) and intake (S pole). It is crucial to emphasize:

- Jets and intakes are not “magnetic charge sources/sinks” and not independent entities;
- They are the enclosure endpoints of the flux tube in space, set by the directionality of boundary conditions.

In static magnetism:

- Jets (N) and intakes (S) are fixed in time;
- The flux tube forms a closed, stable channel between them;
- There is no energy propagating outward or leaking away.

The existence of N and S does not mean that the flux tube is “flowing”; it only shows that the enclosure channel has directional structure.

5.9.3 Double-helix decoupling at the jet (N): release of the outer structure

While the flux tube remains in static equilibrium, nothing propagates at the jet (N). Structurally, however, the jet is the location where outer structure would first be allowed to escape if the enclosure condition were lifted.

We must distinguish carefully:

- It is not the double-helix core that decouples;
- It is the outer phase–energy structure carried by the double helix that decouples.

Once steady enclosure fails at the jet (N):

- The two branches, previously compressed into a single channel, are no longer forced to remain closed;
- They are released along their respective helical directions, forming two decoupled wave-packet structures;
- The “splitting” of the double helix occurs only in the outer structure; the core swirling channel does not break.

Under static conditions, this does not occur; but it defines the structural role of the jet (N).

5.9.4 Recoupling at the intake (S): recombination of the double helix

Conversely, the intake (S) is the re-enclosure end. When the decoupled outer structure once again satisfies enclosure conditions:

- The two double-helix branches are recompressed and locked;

- Their phase relation returns to the steady-state requirement;
- The flux tube becomes a complete enclosure again.

Thus "recombination" does not mean assembling the double helix from scratch; it means that the decoupled outer structure re-enters the double-helix enclosure channel to restore the steady state.

In static magnetism, there is no ongoing decoupling–recoupling process between the jet and intake; they are simply potential structural roles.

5.9.5 Dynamic magnetism: disintegration of a flux tube into propagating packets

When the driving conditions vary in time (e.g. switching, polarity reversal, fast excitation changes), the flux tube can no longer maintain a steady enclosure. What happens is not "a vibrating tube", but rather:

- The flux tube as a whole loses its enclosure condition;
- The double-helix channel is no longer maintained;
- The outer structure is continuously released at the jet (N) and cannot be re-enclosed at the intake (S).

As a result, the flux tube disintegrates into propagating wave packets.

These packets have definite phase and swirling character and can leave the original channel to propagate through space. This is the ontological origin of electromagnetic waves.

We can therefore clearly distinguish:

- Static magnetism: the flux tube exists as a complete double-helix enclosure and does not propagate;
- Dynamic magnetism: the flux tube becomes unstable and disintegrates into packets, forming propagating electromagnetic waves.

5.9.6 Summary

1. Static magnetism corresponds to the stable enclosure state of a double-helix flux tube;
2. Jets (N) and intakes (S) are enclosure endpoints, not sources or sinks;
3. At the jet (N), the outer double-helix structure may decouple; at the intake (S), it may be re-enclosed;
4. Only when the tube as a whole loses enclosure and disintegrates into packets do dynamic magnetism and electromagnetic waves arise;
5. The propagating fields of classical electromagnetism correspond ontologically to the wave-packet state after flux-tube disintegration.

Static magnetism is not "a wave that happens not to move", and dynamic magnetism is not "rotating static magnetism"; they are ontologically different: flux-tube enclosure state vs. its wave-packet disintegration.

5.10 Chapter summary

This chapter offers a complete "enclosure universe" interpretation of the magnetic field:

1. The magnetic field is an Ω-enclosure flow tube;
2. Flux tubes necessarily form under the $|g| = c$ constraint;
3. Jets and intakes are topological defects of the enclosure boundary;
4. The absence of magnetic monopoles follows from $\nabla \cdot (\nabla \times g) = 0$;
5. Stable flux tubes obey an $8\pi k$ closure rhythm;
6. The difference between magnetism and light is whether ϕ is locked or released.

Transition to the next chapter

The next chapter introduces the frequency tree: Why does the same "8" keep reappearing in electrons, flux tubes, black holes, and the solar system?

We will rigorously distinguish:

- the frequency chain,
- the enclosure chain, and
- the harmonic chain,

and show that the progression 8→8k→8π is not an analogy, but the inevitable result of an upgraded enclosure dimension.

Chapter 6 The Frequency Tree and Octupling

— Strictly Separating the Three Chains: Frequency / Enclosure / Harmonic (Avoid All Mixing)

6.1 A "no-mistake axiom": the three chains must never be interchanged

In previous chapters we encountered three seemingly similar—but physically distinct—quantities:

- the electron's 8-event mother number;
- the flux tube's 8k closure rhythm;
- the black hole's 8π enclosure invariant.

If we do not first clarify where these 8's come from and which layer they belong to, the theory immediately collapses into confusion. So this chapter begins by stating a "no-mistake axiom".

Theorem 6.1 (Three-Chain Separation Theorem)

In the enclosed-universe picture, all discussions of scale and frequency must strictly distinguish three chains: the frequency chain, the enclosure chain, and the harmonic chain.

Any derivation that mixes these three is guaranteed to produce dimensional or physical errors. We now fix them one by one.

6.2 The frequency chain (first-order dimension): length–frequency duality

The frequency chain answers exactly one question:

Given a spatial scale ℓ, what is the minimal geometric rotation frequency?

The answer follows directly from the composite-light-speed axiom $|g| = c$.

In any stable structure where the tangential component dominates and the radial component is suppressed,

$$\Omega(\ell) \sim \frac{c}{\ell} \tag{6.1}$$

This is a purely geometric duality:

- smaller scale → higher rotation frequency;
- larger scale → lower rotation frequency.

It involves no quantum, no action, no closure counting.
It is the "trunk" of the entire frequency tree.

6.2.1 Three typical nodes on the frequency chain

- Electron enclosure scale ℓ_e: $\Omega_e \sim \frac{c}{\ell_e}$
- Flux tube scale R: $\Omega_{tube} \sim \frac{c}{R}$
- Black-hole shell radius r_s: $\Omega_{BH} \sim \frac{c}{r_s}$

All of these lie on the same frequency chain; only the scale differs.

6.3 The enclosure chain (second-order dimension): geometric origin of action

The enclosure chain does not deal with frequency, but with action.
In this paper we use the enclosure–action relation

$$\hbar = \mu_g \, c \, \ell_\Omega^2 \tag{6.2}$$

Here

- ℓ_Ω is not a "particle radius";
- it is the area-root scale whose enclosure action equals $\hbar$.

6.3.1 Space unit SU and spin unit SU_Ω

Definitions:

- Space unit SU: base length ℓ_0, with base frequency $\Omega_0 = \frac{c}{\ell_0}$;
- Spin unit SU_Ω: scale ℓ_Ω after area enclosure.

Define the enclosure multiplicity (valid only for areas):

$$N_0 = \left(\frac{\ell_\Omega}{\ell_0}\right)^2 \tag{6.3}$$

Key warning
N_0 is an area multiplicity and must never be used as a frequency multiplier.

6.3.2 How the enclosure chain and frequency chain correctly connect

- Enclosure chain: $\ell_\Omega^2 = N_0\,\ell_0^2$;
- Frequency chain then gives: $\Omega_\Omega = \frac{c}{\ell_\Omega} = \frac{\Omega_0}{\sqrt{N_0}}$ (6.4)

The square root is not a trick; it is a dimensional necessity: frequency is first-order, enclosure is second-order.

6.4 The ontological origin of "octupling": the electron's 8-event mother number

We now address the true origin of "8".

In Chapter 2 we rigorously showed: as the smallest stable enclosure, the electron's closure necessarily involves 8 topological events. This is a geometric closure rhythm, not a fit parameter.

6.4.1 The octupling theorem

The outer shell of the electron, in one full stable closure, must pass through 8 minimal events:

$$\frac{\Omega_e}{c}\oint_{\Gamma_e} \mathrm{d}\,s = 8\pi \tag{6.5}$$

Here, the 8 counts the minimal topological events needed for stable enclosure in one loop; it is not a frequency multiplier.

Theorem 6.2 (Octupling Theorem)
The "8" is not an arbitrary factor, but the topological count of minimal stable enclosure closure. It appears for the first time at the electron level.

6.5 The harmonic chain: discretization by system boundary conditions

The third chain is the easiest to misuse.

The harmonic chain refers to the discrete sequence of allowed modes in a given system (atom, flux tube, solar system, etc.). Its origin is the boundary conditions, not enclosure or action.

6.5.1 Typical examples of the harmonic chain

- Atomic energy levels $n = 1,2,3, \ldots$;
- Cavity modes;
- Orbital resonance ratios;
- Multimode cross-section textures of flux tubes.

The harmonic chain can pick modes, but cannot define scales.

6.6 From 8→8k→8π: dimensional up-grading of a single mother number

We can now unify the "8-series" from the previous chapters.

6.6.1 "8": one-dimensional enclosure closure (electron)

- Object: electron enclosure path;
- Physical quantity: event count;
- Result: 8 minimal closure events.

6.6.2 "8k": one-dimensional flow-tube enclosure (flux tube)

For a flux tube, stability is determined by a dimensionless phase accumulation:

$$\Delta\Phi_{tube} = \oint \frac{\Omega}{c}\, ds \tag{6.6}$$

Stable enclosure requires

$$\Delta\Phi_{tube} = 8\pi k, \quad k \in \mathbb{Z}^{+} \tag{6.7}$$

Here:

- 8 comes from the electron's enclosure mother number;
- k comes from multilayer / multimode enclosure.

6.6.3 "8π": two-dimensional enclosure limit (black hole)

A black hole is not a "bigger flux tube"; it is an up-grade in enclosure dimension:

- Flux tube: 1D flow tube;
- Black hole: 2D spherical shell.

Up-grading the dimension introduces the 4π factor of spherical symmetry. The electron's 8-mother number thus up-grades to

$$M_{\mathrm{BH}}\,\Omega_{\mathrm{BH}} = \frac{\mu_g c}{8\pi} \tag{6.8}$$

which is the 8π enclosure invariant.

6.7 The complete structure of the frequency tree

We can organize the three chains into a single tree:

6.7.1 Frequency chain (trunk)

$$\Omega(\ell) \sim \frac{c}{\ell}$$

6.7.2 Enclosure chain (roots)

$$\hbar = \mu_g c \ell_\Omega^2, \quad N_0 = \left(\frac{\ell_\Omega}{\ell_0}\right)^2$$

6.7.3 Topological closure nodes

$$\frac{\Omega_e}{c}\oint_{\Gamma_e} \mathrm{d}\,s = 8\pi, \quad \oint \frac{\Omega}{c} ds = 8\pi k, \quad M_{\mathrm{BH}}\Omega_{\mathrm{BH}} = \frac{\mu_g c}{8\pi}$$

6.7.4 Macroscopic interfaces (preview)

$$\Omega_{\mathrm{b}g}(r) = \frac{c}{r}, \quad \omega_{orb} = \frac{\Omega_{\mathrm{b}g}}{8k}$$

These will serve as the starting point for Chapters 7 and 8.

6.8 Three hard, discriminative predictions from this chapter

1. The stable transverse modes of a flux tube must follow an 8k rhythm, not arbitrary integers;
2. The geometric factor in the black-hole enclosure invariant must be $1/(8\pi)$;
3. Macroscopic phase-locked structures (e.g. planetary orbits) must exhibit 1/(8k) sub-division, not a continuous distribution.

6.9 Chapter summary

This chapter has built the frequency backbone of the enclosed universe:

- The frequency chain gives the length–frequency duality;
- The enclosure chain gives the geometric origin of action;
- The harmonic chain only chooses discrete modes;

- The 8-mother number is the minimal enclosure closure rhythm;
- 8→8k→8π is the natural result of up-grading the enclosure dimension.

Transition to next chapter
The next chapter moves to the extreme enclosure state—a black hole: Why is a black hole not a singularity but an enclosure shell? Why must jets appear? Why is 8π unavoidable?

Chapter 7 Black Holes: the Extreme State in an Enclosed Universe

— From the Electron's 8 to the Black Hole's 8π: Dimensional Up-grading of Enclosure, No-Singularity Theorem, and Jet Endpoints

7.1 Redefining black holes: not singularities, but enclosure limits

In the enclosed-universe framework, the only physical entity is the space-velocity field g, obeying the composite light-speed axiom

$$|g| = c \tag{7.1}$$

Any description involving "infinite speed" or "infinite curvature" is therefore forbidden at the ontological level.

A black hole must be redefined as follows.

Definition 7.1 (Black hole) ——A black hole is the enclosure limit state of the space-velocity field under spherical symmetry: at its boundary, the radial component is completely suppressed and the tangential composite speed saturates at c.

In other words, a black hole is not the end of falling, but the boundary of enclosure.

7.2 Geometric conditions of the black-hole shell: tangential light-speed saturation

Using the correct decomposition from Chapter 1, at the spherical-shell enclosure limit we have

$$g_r(r_s) \to 0, \quad g_t(r_s) \to c, \quad |g| = c \tag{7.2}$$

where r_s is the black-hole shell radius.

This definition immediately yields three key facts:

1. The event horizon is the enclosure boundary, not a singularity;
2. Shell stability is due to tangential light-speed saturation;
3. No superluminal or divergent quantities are allowed inside.

7.3 Characteristic frequency of a black hole: a direct result of the frequency chain

From Chapter 6, the frequency chain trunk is locked as $\Omega(\ell) \sim \frac{c}{\ell}$.

At the enclosure scale r_s, the black hole's characteristic frequency is

$$\Omega_{BH} \simeq \frac{c}{r_s} \tag{7.3}$$

This is purely geometric—no gravitational field equation is needed.

7.4 Where does 8π come from? Dimensional up-grading of enclosure

This is the core of the chapter.

7.4.1 Reminder: the role of "8" for electrons and flux tubes

- Electron: 1D enclosure path closure with 8 minimal events;
- Flux tube: 1D flow-tube cross-section enclosure with phase closure $8\pi k$.

Common feature: enclosure takes place on a one-dimensional closed object.

7.4.2 Essential difference for black holes: two-dimensional enclosure

A black hole is not a "stronger flux tube"; it is a jump in enclosure dimension:

- Flux tube: 1D flow tube (line × cross-section);
- Black hole: 2D spherical shell.

The minimal geometric fact about 2D enclosure is that spherical symmetry necessarily introduces the solid-angle factor 4π. The minimal enclosure mother number 8 must still be present.

Thus the smallest enclosure factor for a black hole can only be: $8 \longrightarrow 8\pi$

7.5 The black-hole enclosure invariant: the exact form of 8π

Chapter 2 established the enclosure consistency invariant $M\Omega = \mu_g c$

For a black hole as a 2D enclosure limit, this invariant must be renormalized by spherical symmetry, giving

$$M_{BH}\,\Omega_{BH} = \frac{\mu_g\, c}{8\pi} \tag{7.4}$$

This is not a correction term but the inevitable consequence of dimensional up-grading.

7.6 Radius–mass relation for black holes (pure enclosure geometry)

From (7.3) and (7.4) we obtain

$$M_{\mathrm{BH}} \frac{c}{r_s} = \frac{\mu_g c}{8\pi} \Rightarrow r_s = 8\pi \frac{M_{\mathrm{BH}}}{\mu_g} \tag{7.5}$$

This is a pure enclosure–geometric relation:

- no curvature is invoked;
- no gravitational field equation is introduced;
- no singularity assumption is needed.

7.7 No-singularity theorem: a hard constraint of the enclosed universe

Theorem 7.1 (No-Singularity Theorem)
In an enclosed universe obeying $|g| = c$, any enclosure limit state can only form a shell of finite thickness; there is no physical singularity at the center.

Two reasons suffice:

1. The speed magnitude is strictly capped at c;
2. The gradient of the enclosure potential n can be large, but cannot be infinite without destroying enclosure conditions.

Thus a black hole interior is a low-frequency, high-enclosure-density helical cavity, not a singularity.

7.8 Black-hole jets: inevitable result of endpoint breaking

Chapter 5 already showed

$$\nabla \cdot (\nabla \times g) = 0$$

which implies that any non-closed enclosure structure must have endpoints in pairs. The same holds for the black-hole shell.

7.8.1 Geometric origin of jets

When the enclosure shell suffers a local break in some direction:

- g_r is no longer fully suppressed;
- the space flow leaks out along that direction;
- the leakage is still constrained by the right-handedness of Ω.

Thus:

- Jet (N pole): space flow is released outward;
- Intake (S pole): the compensating endpoint.

7.8.2 Complete isomorphism with flux tubes

Object	Enclosure dimension	Endpoint type
Flux tube	1D flow tube	N/S poles (jet/intake)
Black hole	2D spherical shell	Bipolar jets

Jets are not "matter emitted from a singularity"; they are releases from enclosure-boundary endpoints.

7.9 Unified view: 8, 8k, and 8π are the same mother number

We can now place all three levels side by side:

- Electron: 1D enclosure closure →8;
- Flux tube / macroscopic phase-locking: 1D flow-tube enclosure →8k;
- Black hole: 2D spherical shell enclosure →8π.

This is not analogy, but a natural outcome of up-grading the enclosure dimension.

7.10 Chapter summary

This chapter has completed the ontological rewrite of black holes in an enclosed universe:

1. A black hole is an enclosure limit, not a singularity;
2. The event horizon is the enclosure boundary where the tangential speed saturates at c;
3. The black hole's characteristic frequency satisfies $\Omega_{\mathrm{B}H} = c/r_s$;
4. The 8π enclosure invariant follows from dimensional up-grading;
5. The no-singularity theorem follows directly from $|g| = c$ and enclosure saturation;
6. Jets are boundary endpoints of the enclosure, isomorphic to flux-tube endpoints.

Transition to next chapter

The next chapter turns to macroscopic enclosure tests: if the 8-mother number is a genuine geometric mother number, then at the scale of the solar system it must reappear as an "8k phase lock".

We will construct the background rotation $\Omega_{\mathrm{b}g}(r) = \frac{c}{r}$, and derive the discrete structure of planetary orbits together with auditable predictions.

Chapter 8 Solar-System Encapsulation: Macroscopic Phase-Locking Axiom and Orbital Shell Formulae

— from $\Omega_{\mathrm{b}g}(r) = \frac{c}{r}$ to $r \propto k^2, T \propto k^3$

8.1 Aim and Method of This Chapter

This chapter does not do data fitting. It does exactly one thing: push the "8 mother number" encapsulation-closure rhythm up to astronomical scales and derive orbital master formulas that are computable and auditable.

The only allowed inputs (already fixed earlier) are:

1. Composite light-speed axiom: $|g| = c$;
2. Frequency-chain backbone: $\Omega(\ell) \sim c/\ell$;
3. Encapsulation closure mother number: electronic 8→ macroscopic 8k.

Any new hypothesis not contained within these three is strictly forbidden.

8.2 Solar-System Background Flow: Minimal Form of a Macroscopic Right-Handed Helix

8.2.1 Correct background decomposition

On the scale of the solar system, the minimally consistent decomposition of the space-velocity field (cylindrical symmetry) is

$$g_\theta(r) = c \sin\chi(r), \quad g_r(r) \approx 0, \quad |g| = c \tag{8.1}$$

Far from extreme encapsulation regions (e.g. black-hole shells), the tangential component dominates. In the minimal approximation,

$$\sin\chi(r) \approx 1 \quad \Rightarrow \quad g_\theta(r) \approx c \tag{8.2}$$

This does not mean "a body moving at speed c"; it means the background space flow at that scale is tangentially saturated at the speed of light.

8.2.2 Geometric definition of the background rotation frequency

Define the background rotation frequency as the rate of angular phase advance:

$$\Omega_{\mathrm{b}g}(r) \equiv \frac{g_\theta(r)}{r} \tag{8.3}$$

Using (8.2) we obtain the uniquely allowed background frequency:

$$\Omega_{\mathrm{b}g}(r) = \frac{c}{r} \tag{8.4}$$

This is not a gravitational law; it is the direct macroscopic manifestation of the frequency chain.

8.3 Macroscopic Phase-Locking Axiom: Orbits as "8k-Closed" Standing-Wave Nodes

In previous chapters, "8" has been shown to be the minimal encapsulation-closure rhythm.
In macroscopic systems, a stable orbit must satisfy one-cycle closure of the background phase.

Axiom 8.1 (Macroscopic encapsulation phase-locking / 8k-closure)
Planetary orbits are macroscopic encapsulation standing-wave nodes, requiring that the background rotation frequency complete 8k minimal closure cycles in one orbital period:

$$\Delta\Phi_{\mathrm{b}g} = \Omega_{\mathrm{b}g}(r)\,T = 2\pi\,(8k), \quad k \in \mathbb{Z}^{+}. \tag{8.5}$$

Hence the orbital angular frequency is

$$\omega_{orb} = \frac{2\pi}{T} = \frac{\Omega_{\mathrm{b}g}(r)}{8k} \tag{8.6}$$

The factor 8 here is not empirical; it is the electronic encapsulation mother number, inherited directly at macroscopic scale.

8.4 Radius Master Law: Locking Condition ⇒ $r \propto k^2$

In the low-velocity limit, orbital motion still satisfies dynamical consistency (equivalent to a Kepler form):

$$\omega_{orb}^2 = \frac{(GM)_\odot}{r^3}\frac{\Omega_{\mathrm{b}g}(r)}{8k} \tag{8.7}$$

Combining (8.6) and (8.4),

$$\left(\frac{c}{8kr}\right)^2 = \frac{(GM)_\odot}{r^3}$$

Solving gives

$$r_k = \frac{64\,k^2\,(GM)_\odot}{c^2} \tag{8.8}$$

Conclusion 1 (Orbital shell radius law)
Planetary orbital radii must form a discrete shell structure with $r \propto k^2$.
This is not an analogy with atoms; it is the same encapsulation-closure mechanism manifesting at a different scale.

8.5 Period Master Law: Directly Obtaining $T \propto k^3$

From (8.6) and (8.4) we have $T = \frac{2\pi}{\omega_{orb}} = \frac{16\pi\,k\,r}{c}$

Substituting (8.8),

$$T_k = \frac{1024\,\pi\,k^3\,(GM)_\odot}{c^3} \tag{8.9}$$

Conclusion 2 (Period law)
Planetary orbital periods necessarily scale as $T \propto k^3$.
This is the direct result of 8k locking plus the frequency chain.

8.6 k Must Be an Integer: Rounding Rule and Residual Definition

If k is treated as continuous, the model degenerates into pure fitting. We must therefore specify a rounding rule.

8.6.1 Continuously inferred shell index

From an observed semi-major axis r_{obs}, define

$$k_{cont} \equiv \sqrt{\frac{c^2\, r_{obs}}{64\,(GM)_\odot}} \tag{8.10}$$

8.6.2 Minimum-mismatch rounding rule

Definition 8.1 (Integer shell index)

$$k = \mathrm{round}(k_{cont}) \in \mathbb{Z}^+ \tag{8.11}$$

Define the residual

$$\delta k \equiv k_{cont} - k \tag{8.12}$$

In other words:

- Structure = integer k;
- Dynamical / historical correction = δk.

This cleanly separates the structural master law from its correction sources.

8.7 Physical Origin of Errors (Not "Our Calculation Is Bad")

If the master laws (8.8) and (8.9) deviate from observation, the deviations may only come from two sources.

Error source A: background allocation angle not exactly $90°$

More generally,

$$\Omega_{\mathrm{bg}}(r) = \frac{c \sin\chi(r)}{r} \tag{8.13}$$

Then

$$r_k = \frac{64\,k^2\,(GM)\odot}{c^2\,\sin^2\chi}, \quad T_k = \frac{1024\,\pi\,k^3\,(GM)\odot}{c^3\,\sin^3\chi} \tag{8.14}$$

This produces a uniform scale shift, without spoiling the k^2/k^3 structure.

Error source B: multi-body ϕ-interference and migration history

In a multi-planet system, the ϕ-layer develops interference conditions, leading to:

- Some shells being dragged;
- Systematic offsets δk.

These are dynamical corrections, not structural failures.

8.8 Chapter Conclusions

1. The only allowed form of the solar-system background rotation frequency is
$$\Omega_{\mathrm{b}g}(r) = \frac{c}{r}$$
2. Planetary orbits are 8k-encapsulation, phase-locked standing-wave nodes.
3. Radius master law:
$$r_k = \frac{64\, k^2\, (GM)_\odot}{c^2}$$
4. Period master law:
$$T_k = \frac{1024\, \pi\, k^3\, (GM)_\odot}{c^3}$$
5. k must be an integer; δk carries all corrections.
6. Deviations can only come from $\sin\chi \neq 1$ or multi-body interference.

Transition
The next chapter will plug these master formulas directly into real planetary data: no parameter tuning, no curve fitting—simply checking whether the planets really sit on integer shells.
We will present:

- The continuous inverse k_{cont};
- The rounded integer k;
- The residual δk;
- And a structural interpretation of the errors.

Chapter 9 Solar-System Data Audit: Inverting Integer Shell Index k, Rounding, and Error Structure

—from observation to structural quantity: testing 8k encapsulation phase-locking

9.1 Principle: Audit "Structure" Only, No Numerical Fitting

No new models and no tunable parameters are introduced. The only allowed inputs are the master laws already fixed in Chapter 8:

$$r_k = \frac{64\,k^2\,(GM)\odot}{c^2}, \quad T_k = \frac{1024\,\pi\,k^3\,(GM)\odot}{c^3} \tag{9.1}$$

The audit proceeds in four fixed steps:

1. Invert observations to obtain the continuous shell index k_{cont};
2. Apply the minimum-mismatch rule to get integer k;
3. Substitute integer k back into the master laws to predict r_k, T_k;
4. Compare predictions with observations and analyze the structural origin of the residuals.

Any attempt to "improve the fit using fractional k" is deemed a failure.

9.2 Data and Constants (Reproducible)

9.2.1 Observational data

- Planetary semi-major axes r_{obs}: approximate Keplerian elements (AU) from JPL / IAU;
- Planetary periods T_{obs}: public astronomical tables, converted to SI.

9.2.2 Physical constants

- Speed of light: $c = 299792458$ m/s (exact);
- Solar gravitational parameter: $(GM)_\odot = 1.3271244 \times 10^{20}\ m^3/s^2$.

We use $(GM)\odot$ rather than splitting G and $M\odot$ to avoid injecting extra uncertainty.

9.3 Continuous Inversion: Obtaining k_{cont} from Observations

From the radius master law, invert to obtain the continuous shell index:

$$k_{cont} = \sqrt{\frac{c^2\,r_{obs}}{64\,(GM)_\odot}} \tag{9.2}$$

This is a purely observational inverse and is not constrained to be an integer.

9.4 Rounding: Separating Structure from Corrections

9.4.1 Definition of the integer shell index

Definition 9.1 (Integer shell index)

$$k = \mathrm{round}(k_{cont}) \in \mathbb{Z}^+ \tag{9.3}$$

9.4.2 Definition of the residual

$$\delta k \equiv k_{cont} - k \tag{9.4}$$

Key points:

- k: structural quantity (encapsulation-shell label);
- δk: correction container (history of migration, multi-body interference, background deviation, etc.).

Important:
Structure is determined solely by integer k; δk must not be used to modify the master laws.

9.5 Predicting Orbits with Integer k and Comparing to Observations

Using (9.1), compute

$$r_{pred}(k), \quad T_{pred}(k) \tag{9.5}$$

Define relative errors:

$$\varepsilon_r = \frac{r_{pred} - r_{obs}}{r_{obs}}, \quad \varepsilon_T = \frac{T_{pred} - T_{obs}}{T_{obs}} \tag{9.6}$$

9.6 Summary of Audit Results (Representative Planets)

How to read:

- If k_{cont} lies close to some integer k, the planet is near an encapsulation shell;
- If $\varepsilon_r, \varepsilon_T$ have the same sign and similar magnitude, the error comes from a common structural correction rather than model failure.

Planet	$\boldsymbol{k_{cont}}$	**k**	$\boldsymbol{\delta k}$	$\boldsymbol{\varepsilon_r}$	$\boldsymbol{\varepsilon_T}$
Mercury	~ 783	783	-0.21	$\sim 10^{-4}$	$\sim 10^{-4}$
Venus	~ 1070	1070	+0.05	$\sim 10^{-4}$	$\sim 10^{-4}$
Earth	~ 1258	1258	+0.16	$\sim 10^{-4}$	$\sim 10^{-4}$
Mars	~ 1553	1553	+0.01	$\sim 10^{-4}$	$\sim 10^{-4}$
Jupiter	~ 2869	2869	-0.43	$\sim 10^{-3}$	$\sim 10^{-3}$
Saturn	~ 3885	3885	-0.09	$\sim 10^{-4}$	$\sim 10^{-4}$
Uranus	~ 5509	5509	-0.42	$\sim 10^{-3}$	$\sim 10^{-3}$
Neptune	~ 6895	6895	-0.05	$\sim 10^{-3}$	$\sim 10^{-3}$

First-principles conclusion:
All planets have k_{cont} close to some integer k, and using that integer k in the master laws simultaneously predicts both radius and period.

9.7 Physical Interpretation of the Error Structure (Not "Bad Fit")

9.7.1 Same-origin error: $\varepsilon_T/\varepsilon_r = 3/2$

From the two master laws in Chapter 8, logarithmic differentiation yields

$$\frac{\varepsilon_T}{\varepsilon_r} = \frac{3}{2} \tag{9.7}$$

This is a structural criterion:

- If observed errors satisfy the 3/2 ratio, they arise from the same encapsulation master law;
- Significant deviation indicates the need for extra physics.

Overall, planetary data cluster around this ratio, supporting the self-consistency of the master laws.

9.7.2 Error source A: background allocation angle deviations

More general background rotation:

$$\Omega_{\mathrm{b}g}(r) = \frac{c \sin \chi(r)}{r} \tag{9.8}$$

$\sin \chi \neq 1$ induces a uniform multiplicative shift but preserves integer k.

9.7.3 Error source B: multi-body interference and migration history

- Giant planets (especially Jupiter) most strongly rewrite the ϕ-layer;
- Outer planets more easily show systematic δk.

Structurally, Jupiter is closer to an "ideal shell anchor," while deviations of Saturn and the outer planets are more plausibly due to migration and resonances.

9.8 Chapter Conclusions

1. Integer shell index k is the true structural parameter of solar-system orbits.
2. Using integer k predicts radius and period simultaneously.
3. Errors exhibit a same-origin structure $(\varepsilon_T/\varepsilon_r \approx 3/2)$, supporting the encapsulation master laws.
4. δk is the proper container for migration, interference, and historical effects.
5. The solar system is not a random arrangement, but a macroscopic encapsulation standing wave.

Transition

Thus, from electrons, flux tubes, and black holes to the solar system, the same "8 mother number"

reappears over more than seven orders of magnitude and is confirmed by data audits.
The next chapter will complete the final piece: Where do atoms and elements sit in the encapsulated universe, and why must the periodic table exist?

Chapter 10 Atoms and Elements

—from the encapsulated universe to the periodic table: only minimal conclusions and a single channel

10.1 Scope of this chapter

By the end of Chapter 9, this work has completed the main closed loop of the encapsulated-universe picture:

- Microscopic: electron (8 events) → proton ($\pi^2\alpha^2$ threshold) → neutron (ϕ-compensation)
- Field: flux tube (Ω encapsulation) → nozzle (N pole) / intake (S pole) → magnetic vs radiative (light) states
- Extreme: black hole (8π encapsulation)
- Macroscopic: solar system ($8k$-locking) → data audit

In this chapter we no longer engage in "derivation contests." We only do three things:

1. Give a geometric definition of the atom;
2. Give a topological definition of the element;
3. Indicate the structural reason why the periodic table must exist.

More detailed work on spectroscopy, fine / hyperfine structure, the valley of stability, and chemical bonding is deferred to Nature from First Principles 4.

10.2 What is an atom: not "electrons orbiting a nucleus," but the family of closable outer-shell phase solutions

In an encapsulated universe:

- The electron proper is a stable encapsulated state (Chapter 2);
- The nucleus is a background of strong n-encapsulation plus Ω-texture (Chapters 3 and 4).

Therefore an atom should not be defined as "electrons orbiting a nucleus," but rather as:

Definition 10.1 (Atom)

An atom is the family of closable solutions of the electron outer-shell phase ϕ in the nuclear encapsulation background.

Its minimal closure condition is

$$\oint \nabla \phi \cdot d\mathbf{s} = 2\pi m, \quad m \in \mathbb{Z}^+ \tag{10.1}$$

Three clarifications (we only state structure, not detailed calculations):

- Discreteness comes from the closure condition, not from any quantum postulate;
- The "orbit" is not the trajectory of the electron proper, but the set of closure nodes of the outer shell;
- The "probability cloud" is the statistical appearance of the outer-shell structure sampled by encapsulation events.

Minimal derivation of atomic shells: closure + nuclear encapsulation background

Consider the minimal encapsulation background provided by the nucleus at atomic scales. Take the nuclear encapsulation potential to be spherically symmetric:

$$n(r) = n_\infty + \frac{\kappa_n}{r} \tag{10.1.1}$$

where $\kappa_n > 0$ is a nuclear encapsulation-strength parameter with dimensions of length.

By the definition $\mathbf{A}_g = c^2 \nabla n,$ the gravitational / infall acceleration provided by the nucleus is

$$|\mathbf{A}_g| = \frac{c^2 \kappa_n}{r^2} \tag{10.1.2}$$

The closure condition for the outer-shell phase requires

$$\oint \nabla \phi \cdot d\mathbf{s} = 2\pi m, \quad m \in \mathbb{Z}^+ \tag{10.1.3}$$

Combining the frequency chain and the minimal encapsulation closure rhythm (the "8-mother number"), we take the outer-shell tangential drift speed as

$$v_\theta = \frac{c}{8m} \tag{10.1.4}$$

The stable residence radius is obtained by matching centripetal acceleration with the infall field:

$$\frac{v_\theta^2}{r} = \frac{c^2 \kappa_n}{r^2} \tag{10.1.5}$$

Solving gives:

$$r_m = 64\, \kappa_n\, m^2 \tag{10.1.6}$$

This shows that atomic shell radii are discretized as $r \propto m^2$, entirely determined by encapsulation geometry.

10.3 Why atoms are stable: the nucleus is not a "source of force," but a boundary condition

In this framework, the role of the nucleus is not to "attract" electrons, but to provide three boundary facts:

1. Strong n-encapsulation background → sets the radial window in which outer shells can close;
2. Strong Ω locking axis (Z-axis) → sets the directional organization of outer shells;
3. Single-channel visibility → guarantees that the externally visible charge does not drift with internal complexity.

Thus, the fundamental reason "electrons do not fall into the nucleus" is: the nuclear interior region is a non-closable region for the outer shell. The outer shell can only reside at radii where closure is possible.

10.4 What is an element: not a material label, but a topological integer

In Chapters 3 and 4 the charge readout rule was fixed as:

$$q = \Delta\phi\, q_0, \quad \Delta\phi \in \mathbb{Z} \tag{10.2}$$

We showed that:

- The proton outwardly retains a single channel with $|\Delta\phi| = 1$;
- The neutron has $\Delta\phi = 0$ (compensation).

Based on this we give the ontological definition of an element:

Definition 10.2 (Element)
An element is uniquely specified by the "effective number of phase outlets" Z the nucleus presents to the outside,

$$Z \equiv |\Delta\phi_{nucleus}| \in \mathbb{Z}^+$$

Immediate consequences:

- Z is a topological integer, not an empirical label;
- The neutron number N changes the encapsulation capacity and stability of the nucleus but does not change Z;
- Hence isotopes arise naturally.

10.5 Why neutrons are necessary for the world of elements to exist

Chapter 4 established the structural fact that the neutron is a de-charged encapsulated state: a proton core plus a ϕ-compensation shell. At the level of elements, the interface conclusion is: neutrons allow the nucleus to increase its n and Ω capacity without increasing the external ϕ load.

Without neutrons:

- Multiple protons would overload the phase outlets;
- The encapsulation would be unstable;
- Structure would terminate at hydrogen.

Therefore: no neutrons, no periodic table.

10.6 Interface between atomic spectra and the frequency tree

Chapter 6 established the backbone of the frequency tree:

- Frequency chain: $\Omega \sim c/\ell$
- Encapsulation chain: $\hbar = \mu_g c \ell_\Omega^2$
- Topological closure: $8 \to 8k \to 8\pi$

In this language: The atomic spectrum = structural difference-frequencies between distinct outer-shell closure solutions.

Its intensity and compression rules:

- Inherit the electron's 8-mother number;
- Are projected through the nucleus's single-channel visibility;
- Are further compressed at the macroscopic level by factors involving α (details in Nature from First Principles 4).

10.7 Technical "ceiling" of this book: three layers, three chains, one ontology

Without introducing any new ontological entities, this work has completed the following closure:

- Single ontology: the space-velocity field g;
- Three-layer structure: ϕ, Ω, n;
- Three chains: frequency chain / encapsulation chain / harmonic chain;
- Three validations: electron → black hole → solar system.

Within this framework:

- Atoms are not quantum oddities;
- Elements are not merely empirical categories;
- The periodic table is not a coincidental arrangement.

They are inevitable mid-scale manifestations of the encapsulated universe.

10.8 Closing statement

The universe is not a particle universe, nor a wave universe, but an encapsulated universe. Electrons, flux tubes, black holes, planets, atoms, and elements are simply stable solutions of the same space-velocity field at different encapsulation dimensions and scales.

Nature from First Principles 3 thus completes the unified closure of the "encapsulated universe." More detailed work on the valley of nuclear stability, periodic-table structure, chemical bonds, and spectral compression will use this book as the sole geometric entry point and will be developed systematically in Nature from First Principles 4.

10.9 Atomic structure from the single-channel and flux-tube viewpoint

In the preceding sections of Chapter 10, we gave interface-level definitions of atoms and elements from the lowest-order viewpoint of phase closure (ϕ):

- Atoms are described as the family of closable phase solutions of the electron outer shell in the nuclear encapsulation background;
- Elements are topologically labeled by the number of effective phase outlets Z that the nucleus presents to the outside.

This is logically self-consistent and provides the necessary lowest-order interface, but it is not the final characterization of atomic structure.

As the encapsulated-universe framework is pushed further, it becomes clear that:

Stable encapsulated states do not have many independent external channels; they are strictly single-channel systems.

From this higher-level vantage point, three structural facts become unavoidable.

(1) Single-channel principle: a stable encapsulated state allows only one physical channel to the outside

At both microscopic and macroscopic levels, any encapsulated state that can exist stably over long times obeys: it has only one effective physical channel to the outside.

- Electron: couples outward via its surrounding flux-tube structure;
- Proton: manifests externally via a single topological outlet and its surrounding flux tube;
- Black hole: manifests via a finite number of broken boundary ports, forming bipolar jets;
- Atom: couples and exchanges energy via a single coaxial channel between nucleus and electron.

Thus an atom is not a "multi-degree-of-freedom superposition system," but a single-channel encapsulation system.

(2) Electrons and protons are not "bare encapsulated states," but encapsulation units carrying surrounding flux tubes

Earlier in the book, flux tubes, as Ω-encapsulated flow tubes, were established as basic structures. At a more refined level:

- An electron that exists stably necessarily carries its own surrounding flux-tube structure;
- A proton, as a deeper encapsulation, also has a corresponding surrounding flux tube;
- These flux tubes are not add-ons but the only way encapsulated states manifest to the outside.

At the atomic scale, electrons and nuclei do not interact by "superposed force fields," but by coaxial alignment, phase-locking, and standing-wave conditions along a single flux tube.

(3) A new level of understanding of atomic stability: standing waves in a channel, not orbital motion

From the single-channel / flux-tube viewpoint, atomic stability no longer arises from "electrons orbiting the nucleus," nor merely from "phase-closure conditions," but from:

The electron and the nucleus forming stable standing-wave matching within a single shared flux-tube channel.

In this framework, "atomic energy levels" correspond to:

- Allowed standing-wave modes in the single channel;
- Determined jointly by channel length, encapsulation conditions, and topological closure;
- Independent of the traditional orbital picture.

This naturally ties together:

- The discreteness of atomic spectral lines;
- Selection rules for emission and absorption;
- The appearance of light as a decoupled propagating state in the channel.

(4) Interface statement: this book stops here; higher-order structure will be developed later

We emphasize that this single-channel and flux-tube atomic model belongs to a higher-order structural level than what was fully developed in this book.

The goal of Nature from First Principles 3 is to complete the geometric and topological closure proof of the "encapsulated universe." At the level of atoms and elements, we therefore provide only the lowest-order interface description based on ϕ-closure.

In particular:

- A systematic derivation of the single-channel atomic model;
- The structure of atomic spectra under flux-tube coupling;
- A unified explanation of electromagnetic waves, radiation, and absorption as channel propagation states

will be core topics of the next stage of work, and will be developed systematically in Nature from First Principles 4.

Summary of this subsection

This subsection introduces no new assumptions. It simply distinguishes two levels:

- The level completed in this book:
  Atoms as lowest-order encapsulation interfaces defined by phase closure;
- The higher-order level already identified but not elaborated here:
  Atoms as single-channel, flux-tube–coupled encapsulation systems.

This separation of levels preserves the logical completeness of Nature from First Principles 3 while providing a clear and robust interface for the natural extension of the theory.

Appendix A

Discriminating formulas and consistency audits for the encapsulated universe

A.1 Single axiom and normalization constraint

A.1.1 Composite light-speed axiom (root of the entire book)

$$\boxed{|g(\mathbf{x},t)| = c} \tag{A1}$$

- Physical status: sole axiom;
- Direct consequence: any helical, swirling, or encapsulated structure may only change direction, not speed magnitude;
- Falsification criterion: if any stable structure requires $|g| > c$, the framework self-destructs.

A.1.2 Correct helical decomposition

$$g_\theta = c\sin\chi, \quad g_z = c\cos\chi, \quad g_r \to 0 \tag{A2}$$

- Role: enforces $|g| > c$
- Common error: writing $g_\theta \sim cf(r)$, $g_z \sim ch(r)$ directly
- Diagnostic: whether a distribution angle χ is explicitly introduced

A.2 Three-layer structure and three field definitions (dimension locked)

A.2.1 Three-layer structure (minimal state space)

$$\phi \in S^1, \quad \Omega = |\nabla \times g|, \; n = \frac{1}{c}\nabla \cdot g \tag{A3}$$

- ϕ: direction / phase (electricity, information);
- Ω: swirling strength (magnetism, spin);
- n: encapsulation potential (objecthood, mass).

A.2.2 Three geometric fields (not mutually substitutable)

$$\mathbf{E}_g = \nabla\phi, \quad \mathbf{B}_g = \nabla \times g, \quad \mathbf{A}_g = c^2\nabla n \tag{A4}$$

- $\mathbf{A}_g$ naturally has dimensions of acceleration;
- It is not allowed to call ∇n directly "gravity."

A.3 Encapsulation consistency invariants (from micro to macro)

A.3.1 Generic encapsulated states (electron, proton, magnetic port)

$$\boxed{M\Omega = \mu_g c} \tag{A5}$$

- First appears: electron encapsulation (Chapter 2);
- Uses:
  - universality of charge;
  - spin–mass duality;
- Falsification: observing a stable encapsulated state that violates this product.

A.3.2 Black-hole encapsulation limit

$$\boxed{M_{\mathrm{B}H}\Omega_{\mathrm{B}H} = \frac{\mu_g c}{8\pi}} \tag{A6}$$

- Origin: upgrade from 1D encapsulation → 2D spherical shell;
- Key point: 8π is not a fit parameter, but a dimensional-upgrade factor;
- Falsification: a stable black-hole limit that does not obey this ratio.

A.4 The "8" mother number and its three manifestations

A.4.1 Electron: 8-event topological mother number

$$\boxed{N_{event} = 8} \tag{A7}$$

- Origin: double-branch phase + directional closure;

- Physical identity: minimal topological count for stable encapsulation;
- Non-adjustable: not an empirical constant.

A.4.2 Flux tubes: 8(k) closure quantization

$$\oint \frac{\Omega}{c}\, ds = 8\pi k, \quad k \in \mathbb{Z}^+ \tag{A8}$$

- Note: it must be Ω/c, otherwise dimensions are wrong;
- Meaning: phase closure for 1D flow-tube encapsulation;
- Experimental interfaces: magnetic segment patterns, duct modes.

A.4.3 Black holes: 8π encapsulation invariant

See (A6).

Unified relation of the three:

$$\boxed{8 \longrightarrow 8k \longrightarrow 8\pi} \tag{A9}$$

corresponding to: 1D path → 1D flow tube → 2D spherical shell.

A.5 Frequency tree: strict separation of three chains

A.5.1 Frequency chain (first-order dimension)

$$\boxed{\Omega(\ell) \sim \frac{c}{\ell}} \tag{A10}$$

- Controls: scale ↔ rotation frequency;
- Applies to: electrons, flux tubes, black holes, solar-system background.

A.5.2 Encapsulation chain (second-order dimension)

$$\boxed{\hbar = \mu_g c \ell_\Omega^2}, \quad \boxed{N_0 = \left(\frac{\ell_\Omega}{\ell_0}\right)^2} \tag{A11}$$

- N_0 may only be used as an area factor;
- It is forbidden to treat N_0 as a frequency multiplier.

A.5.3 Eight-fold frequency theorem

$$\boxed{\frac{\Omega_e}{c} \oint_{\Gamma_e} \mathrm{d}\, s = 8\pi} \tag{A12}$$

- The "8" comes from topological closure;
- It is not an arbitrary frequency multiple.

A.6 Macroscopic discriminants for the solar system

A.6.1 Background rotation frequency (only allowed form)

$$\boxed{\Omega_{\mathrm{b}g}(r) = \frac{c}{r}} \tag{A13}$$

A.6.2 $8k$ locking condition

$$\boxed{\omega_{orb} = \frac{\Omega_{\mathrm{b}g}}{8k}} \tag{A14}$$

A.6.3 Orbital master formulas (computable)

$$r_k = \frac{64\,k^2\,(GM)\odot}{c^2}, \quad T_k = \frac{1024\,\pi\,k^3\,(GM)\odot}{c^3} \tag{A15}$$

A.6.4 Rounding rule (key to avoiding overfitting)

$$k_{cont} = \sqrt{\frac{c^2 r_{obs}}{64(GM)\odot}}, \quad k = \mathrm{round}(kcont) \tag{A16}$$

- Structural quantity: integer k
- Correction container: $\delta k = k_{cont} - k$

A.7 Error co-origin criterion

From the logarithmic derivatives of (A15) one obtains:

$$\frac{\varepsilon_T}{\varepsilon_r} = \frac{3}{2} \tag{A17}$$

- Meaning: radial and period errors have the same structural origin;
- Falsification: systematic deviation from the 3/2 ratio.

A.8 Summary of this appendix

All testable content of the encapsulated-universe framework ultimately relies only on:

- $|g| = c$;
- The three-layer structure $\phi - \Omega - n$;
- The “8” topological closure mother number;
- And three strictly separated chains (frequency / encapsulation / harmonics).

If “8” does not hold, the framework collapses;
If “8” does hold, cross-scale consistency becomes unavoidable.

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